

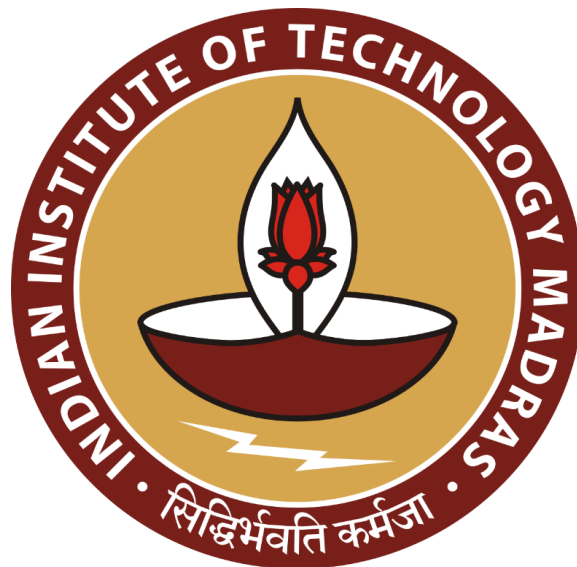
Optimizing Inventory Management and Product Profitability for Astha RCM PUC

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**Optimizing Inventory Management and Product Profitability for Astha RCM PUC**”. I extend my appreciation to **Astha RCM PUC**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Nivedita Prasad

Signature of Candidate: **(Digital Signature)**

Name : Nivedita Prasad

Date : 01-10-2024

1 Executive Summary and Title

Astha RCM PUC, established in 2021, is a department store in B2C retail market sector, in Bero, Ranchi(Jharkhand) which specializes in offering a wide range of products such as groceries, personal care products, apparel, and health supplements. The primary objective of the store is to provide a convenient shopping experience by offering a wide variety of products under one roof. However, the store faces challenges in managing inventory and analyzing profitability.

The main issues are overstocking, which ties up capital, and stockouts, leading to missed sales opportunities. Additionally, the shop lacks visibility into its most profitable products, resulting in inefficient reordering and marketing decisions.

This project proposes to solve these issues by optimizing inventory and doing a profitability analysis. Using sales and inventory data, the project will apply demand forecasting techniques, calculate Economic Order Quantity (EOQ) and reorder points, implement ABC analysis to minimize stockouts and overstocking, and also conduct product profitability analysis to identify the top ten products based on profit and sales volume, to enhance business efficiency.

The expected outcome is optimized inventory management, increased sales of high-profit items, and better space utilization, leading to greater profitability and operational efficiency for Astha RCM PUC.

2 Organization Background

Astha RCM PUC is a department store located in Bero, Ranchi(Jharkhand) which specializes in the sales of a wide category of products such as food, grocery, personal care, health supplements, apparel, footwear and more.

This business operates in the B2C retail market sector and their products come from a single specific brand. The primary goal of this retail business is to provide a convenient shopping experience by offering a variety of goods under one roof.

This retail store was established in October 2021 under the sole proprietorship of Mrs. Sarika Devi. Within a few years, it has built a strong brand identity and gained customer loyalty through its quality products.

The owner herself manages the store along with a skilled worker who even performs accounting and managerial duties of the shop in the absence of the owner.

3 Problem Statement

This project aims to assist Astha RCM PUC in identifying and overcoming its present issues by applying business data management techniques. There will be two primary objectives:

3.1 Inventory Optimization: Minimizing stockouts and overstock of the products and improving overall inventory optimization by analyzing sales pattern and inventory data.

3.2 Product Profitability Analysis: Determining the top ten products by profit and sales volume to focus on pricing and marketing strategies.

4 Background of the Problem

The shop owner mentioned frequent occurrences of being overstocked with certain products of apparel or footwear category and stockout of some other products of health supplements category. Stockouts lead to lost sales and customer dissatisfaction while overstocking ties up capital that could be utilized elsewhere in the business and also leads to wasted shelf space and sometimes potential spoilage, especially for perishable goods.

Additionally, the owner does not have visibility into which products generate the most profit and the profit margins of certain products. Some items sell well but contribute little profit due to low margins. The shop also struggles with accurately predicting customer demand, leading to either overordering or underordering.

Prioritizing products for reordering, marketing, or promotions is also a challenge, as the shop is unsure which items contribute most to profitability or sales volume.

As a small business, Astha RCM PUC does not have the necessary tools or expertise to analyze sales data, forecast demand and manage inventory effectively and they rely on manual tracking and basic accounting methods, which makes it difficult to make informed decisions about stock levels and product focus.

5 Problem Solving Approach

Data Collection:

For inventory optimization, we will collect weekly or monthly sales data of 5-6 months for each product, current inventory data which are stock levels, product categories, and reorder frequency. and, for profitability analysis, we will gather sales volume, selling price, and cost price for each product.

We will use Excel Pivot tables to analyze the data.

Sales Pattern Analysis:

We will use line graphs in Excel to plot sales trends over a time period and identify high-demand periods and low-demand periods and also calculate average sales per week and standard deviations to estimate demand fluctuations.

Demand Forecasting:

Using sales data, we will forecast future demand by calculating moving average and use a line graph to visualize the forecasted demand.

Inventory Optimization:

We will calculate the safety stock level for each product based on its sales and lead time and use Economic Order Quantity (EOQ) or Reorder Point formulas to determine optimal restocking levels.

We will also implement ABC analysis to balance stockout and overstocking as follows:

A items: High sales value, low volume, the products which are critical to keep in stock.

B items: Moderate sales value and volume, the products that need to be regularly monitored and stocked.

C items: Low sales value, high volume, the products that are at risk of overstocking.

We will use a bar chart to display stock levels of product categories to visualize where overstock or stockouts are occurring.

Top Ten Products by Profit(High-Margin Products):

We will compute the total profit by multiplying the profit per unit by the number of units sold for each product.

Then, we will arrange the products in descending order according to their total profit to get the top

ten profitable products.

Top Ten Products by Sales Volume(High-Sales, Lower-Margin Products):

We will compute sales volume, which is simply the total number of units sold for each product and then we will arrange all the products in descending order according to the number of units sold to get the top ten products.

We will use Pareto charts to visualize the top ten products and a Pie chart to display the share of each product category to the total profit.

Marketing and Pricing Strategies:

For high-profit, low-volume products, we can increase sales through targeted marketing e.g., bundle offers, seasonal or bulk-purchase discounts.

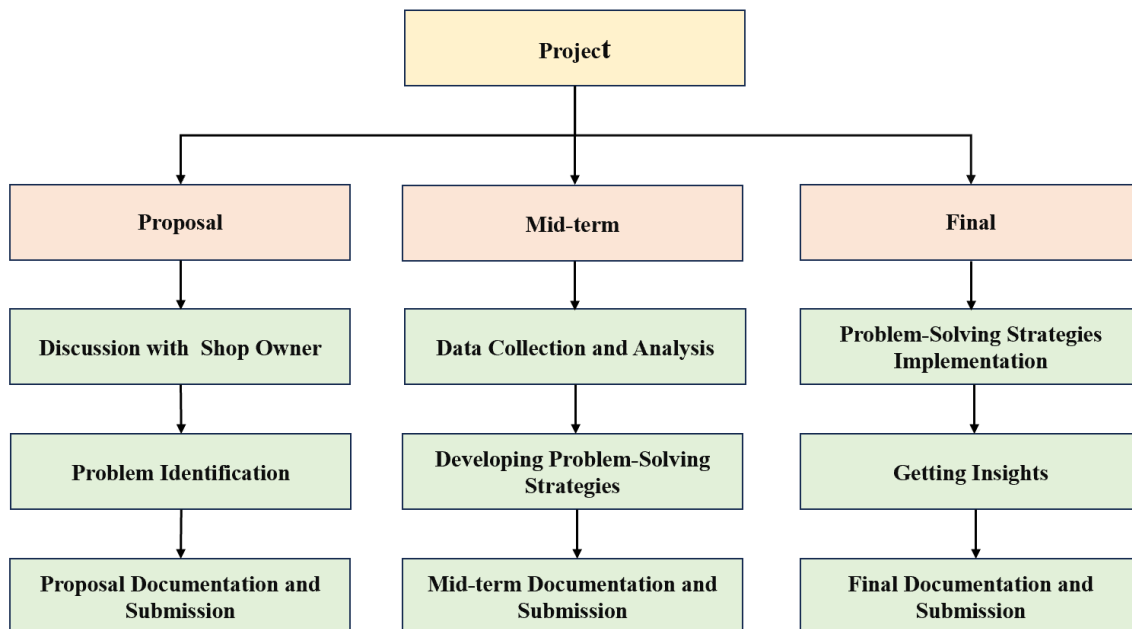
For products with high sales volume but low profit margins, we can adjust the pricing according to the price elasticity of demand to increase profitability.

Using a heat map, we will highlight products with high sales volume, low-profit margins or high-profit, low sales volume products.

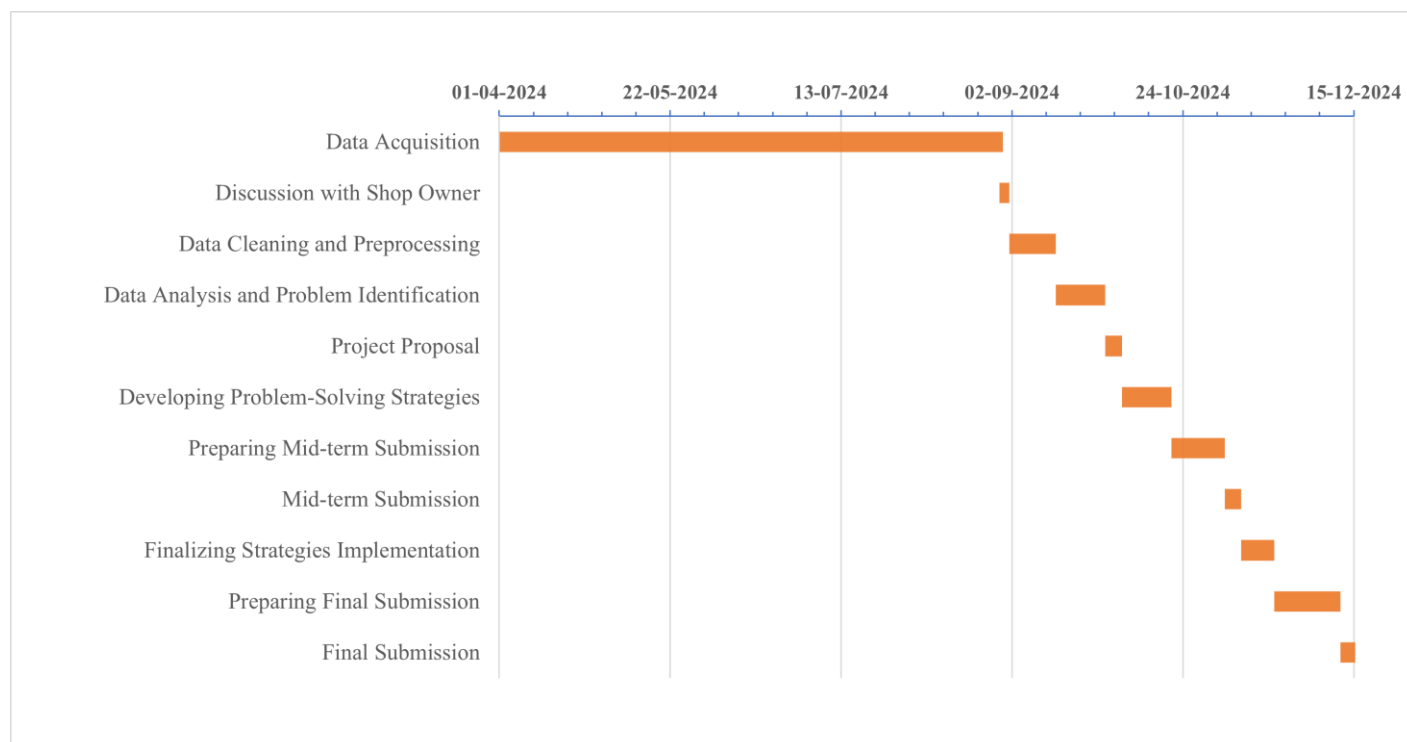
6 Expected Timeline

Tasks	Start	End
Data Acquisition	01-04-2024	30-08-2024
Discussion with Shop Owner	30-08-2024	01-09-2024
Data Cleaning and Preprocessing	02-09-2024	15-09-2024
Data Analysis and Problem Identification	16-09-2024	30-09-2024
Project Proposal	01-10-2024	05-10-2024
Developing Problem-Solving Strategies	06-10-2024	20-10-2024
Preparing Mid-term Submissions	21-10-2024	05-11-2024
Mid-term Submissions	06-11-2024	10-11-2024
Finalizing Strategies Implementation	11-11-2024	20-11-2024
Preparing Final Submissions	21-11-2024	10-12-2024
Final Submissions	11-12-2024	15-12-2024

6.1 Work Breakdown Structure:



6.2 Gantt Chart:



7 Expected Outcome

Reduced Stockouts or Overstocking:

Due to improved demand forecasting and restocking strategies, there will be fewer instances of lost sales opportunities or capital tied up in excess inventory and risks of product spoilage for perishable items.

Better Inventory Management:

Important products (A items) will get more focused attention in managing inventory, ensuring they are always in stock and

by reducing the stock of less important items (C items), we can free up capital and resources to invest in more profitable products, that is, keeping shelves stocked with in-demand products will enhance customer satisfaction and also encourage repeat purchases.

Optimized Reordering Process:

Implementation of reorder points and safety stock levels will ensure timely restocking without over-ordering.

Improved Sales of High-Profit Items:

Focused marketing strategies such as cross-promotions of two different categories of products, bundling a pair of products, or seasonal promotions and discounts on large-quantity purchases for high-profit products may boost their sales volumes.

Data-Driven Pricing Strategies:

Implementing price elasticity analysis will help maintain optimal price levels, avoiding unnecessary discounting or underpricing.

Better Space Utilization:

There will be reduced shelf space wastage by adjusting stock levels to align with demand patterns and increased space for top-selling and profitable products, leading to better space utilization.